

1D CNN — Architecture Variant Comparison

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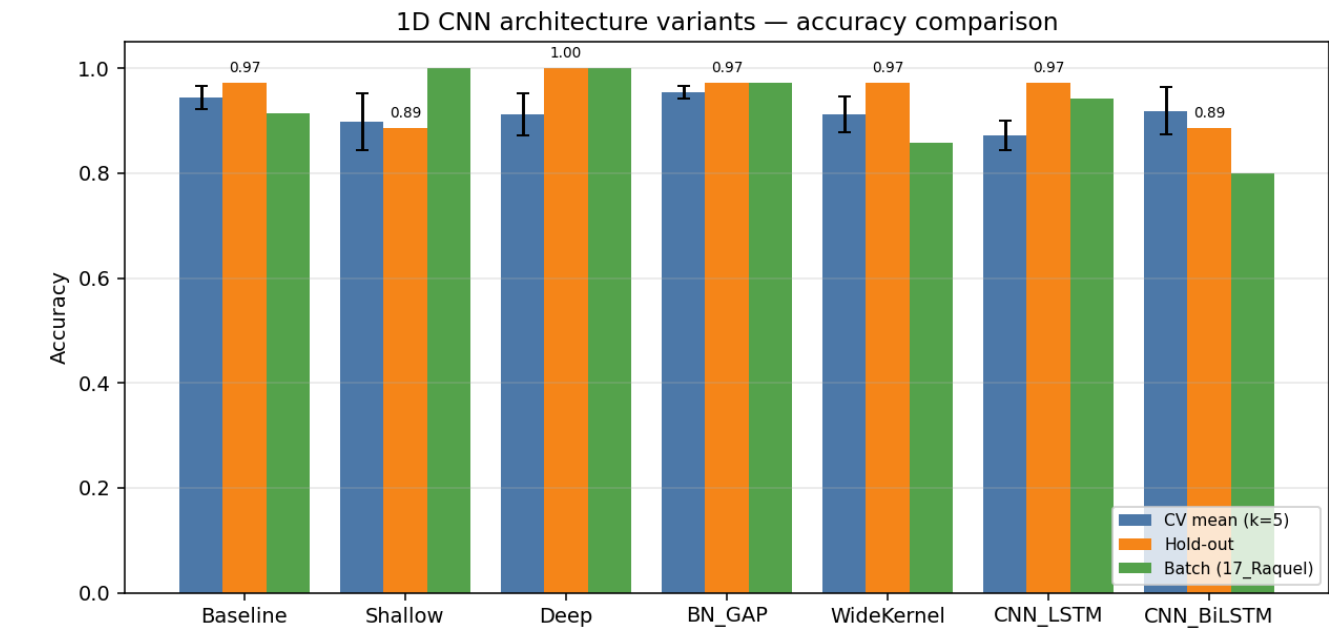
Data & configuration

Training path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
Training path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
Training path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
Training path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
Training path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
Training path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
Training path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
Training path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
Training path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
Training path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
Batch-test root	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
Total trials	341 (train pool 306, hold-out 35)
Classes	7: Double_Lower, Double_Nothing, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 90)
Resample / filter	90 steps · Butterworth cutoff 15.0 Hz, order 4
Normalisation	minmax
CV folds · shuffle	5 · True
Augmentation	on · copies=5 · amplitude_scale, baseline_offset
Epochs · batch size	15 · 16

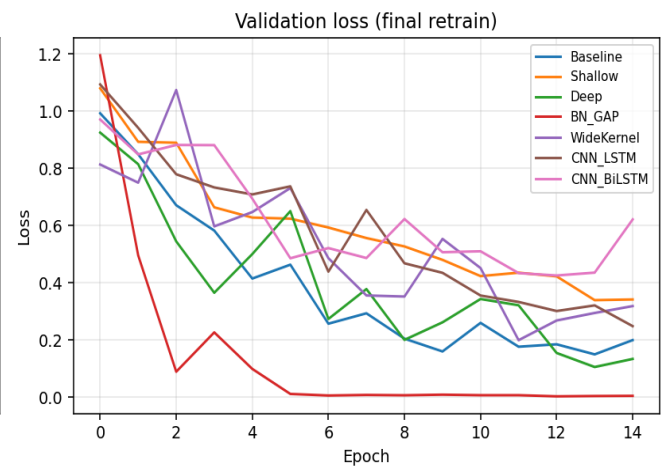
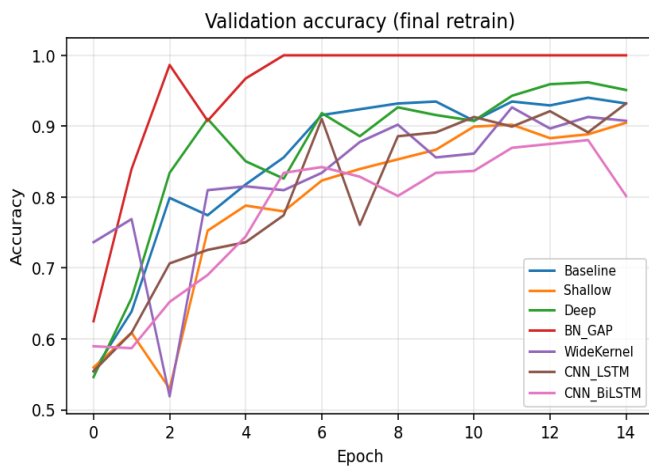
Variant comparison (summary)

Variant	Params	CV mean (k=5)	CV std	Hold-out acc	Hold-out loss	Batch acc	Batch n
Baseline	113,255	0.9445	0.0221	0.9714	0.2050	0.9143	32/35
Shallow	72,487	0.8988	0.0540	0.8857	0.2591	1.0000	35/35
Deep	204,007	0.9117	0.0397	1.0000	0.0289	1.0000	35/35
BN_GAP	57,319	0.9543	0.0122	0.9714	0.1230	0.9714	34/35
WideKernel	131,687	0.9118	0.0337	0.9714	0.0576	0.8571	30/35
CNN_LSTM	68,455	0.8726	0.0282	0.9714	0.0946	0.9429	33/35
CNN_BiLSTM	105,575	0.9184	0.0449	0.8857	0.3652	0.8000	28/35

Best on hold-out: **Deep** · Best on CV: **BN_GAP**



Training curves (final retrain on full training pool)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	113,255
CV mean ± std	0.9445 ± 0.0221
Hold-out test acc / loss	0.9714 · 0.2050
Batch-test acc	0.9143 (32/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9355	0.9355	0.9355
2	1470	61	0.9344	0.9508	0.9344
3	1470	61	0.9836	1.0000	0.9836
4	1470	61	0.9508	1.0000	0.9508
5	1470	61	0.9180	0.9344	0.9180

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	2	5	0.400
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	32	35	0.914

Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	72,487
CV mean ± std	0.8988 ± 0.0540
Hold-out test acc / loss	0.8857 · 0.2591
Batch-test acc	1.0000 (35/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.8710	0.8710	0.8710
2	1470	61	0.9016	0.9016	0.9016
3	1470	61	0.9836	1.0000	0.9836
4	1470	61	0.8197	0.8852	0.8197
5	1470	61	0.9180	0.9180	0.9180

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	0.667	0.400	0.500	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	0.714	1.000	0.833	5
Drum_Roll	1.000	0.800	0.889	5
macro avg	0.888	0.886	0.876	35
weighted avg	0.888	0.886	0.876	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	35	35	1.000

Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	204,007
CV mean ± std	0.9117 ± 0.0397
Hold-out test acc / loss	1.0000 · 0.0289
Batch-test acc	1.0000 (35/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9194	0.9194	0.9194
2	1470	61	0.8361	0.9836	0.8361
3	1470	61	0.9508	0.9836	0.9508
4	1470	61	0.9180	0.9672	0.9180
5	1470	61	0.9344	0.9508	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	1.000	1.000	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	1.000	1.000	1.000	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	1.000	1.000	1.000	35
weighted avg	1.000	1.000	1.000	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	35	35	1.000

Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,319
CV mean ± std	0.9543 ± 0.0122
Hold-out test acc / loss	0.9714 · 0.1230
Batch-test acc	0.9714 (34/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9516	0.9677	0.9516
2	1470	61	0.9672	0.9672	0.9672
3	1470	61	0.9508	0.9836	0.9508
4	1470	61	0.9672	0.9836	0.9672
5	1470	61	0.9344	0.9508	0.9344

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	4	5	0.800
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	5	5	1.000
Overall	34	35	0.971

Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	131,687
CV mean ± std	0.9118 ± 0.0337
Hold-out test acc / loss	0.9714 · 0.0576
Batch-test acc	0.8571 (30/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.9032	0.9194	0.9032
2	1470	61	0.8525	0.9344	0.8525
3	1470	61	0.9508	0.9672	0.9508
4	1470	61	0.9344	0.9672	0.9344
5	1470	61	0.9180	0.9344	0.9180

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	2	5	0.400
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	3	5	0.600
Overall	30	35	0.857

Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,455
CV mean ± std	0.8726 ± 0.0282
Hold-out test acc / loss	0.9714 · 0.0946
Batch-test acc	0.9429 (33/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.8710	0.8871	0.8710
2	1470	61	0.8852	0.9344	0.8852
3	1470	61	0.8525	0.9016	0.8525
4	1470	61	0.9180	0.9180	0.9180
5	1470	61	0.8361	0.9180	0.8361

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.833	1.000	0.909	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	1.000	1.000	5
macro avg	0.976	0.971	0.971	35
weighted avg	0.976	0.971	0.971	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	5	5	1.000
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	3	5	0.600
Overall	33	35	0.943

Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,575
CV mean ± std	0.9184 ± 0.0449
Hold-out test acc / loss	0.8857 · 0.3652
Batch-test acc	0.8000 (28/35)

Cross-validation folds

Fold	Train (post-aug)	Val	Val acc	Best val	Final val
1	1464	62	0.8871	0.9194	0.8871
2	1470	61	0.9180	0.9508	0.9180
3	1470	61	0.9672	0.9836	0.9672
4	1470	61	0.9672	0.9672	0.9672
5	1470	61	0.8525	0.9344	0.8525

Per-class hold-out metrics

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	5
Double_Nothing	1.000	0.800	0.889	5
Double_Pistol_Recoil	1.000	1.000	1.000	5
Double_Raise	1.000	1.000	1.000	5
Double_Wiggle	0.556	1.000	0.714	5
Double_cmere	1.000	1.000	1.000	5
Drum_Roll	1.000	0.400	0.571	5
macro avg	0.937	0.886	0.882	35
weighted avg	0.937	0.886	0.882	35

Batch test per class (17_Raquel)

Class	Correct	Total	Accuracy
Double_Lower	5	5	1.000
Double_Nothing	3	5	0.600
Double_Pistol_Recoil	5	5	1.000
Double_Raise	5	5	1.000
Double_Wiggle	5	5	1.000
Double_cmere	5	5	1.000
Drum_Roll	0	5	0.000
Overall	28	35	0.800